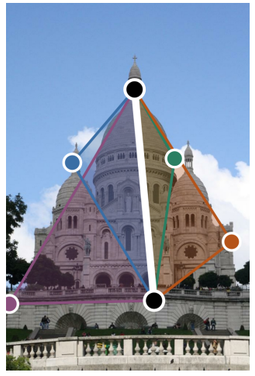
The mathematics of CAPRI

Everything lives on a pencil of planes.

Brackets $[ijkl] = \det(p_i,\ p_j,\ p_k,\ p_l)$ of n points in $\mathbb{P}^3$. Gauge: $p_k \mapsto \lambda_k\ A\ p_k$ for any frame A and any scales λ_k.

a Invariants



Four planes through the line $p_i p_j$, one through each of p_c, p_d, p_e, p_f:

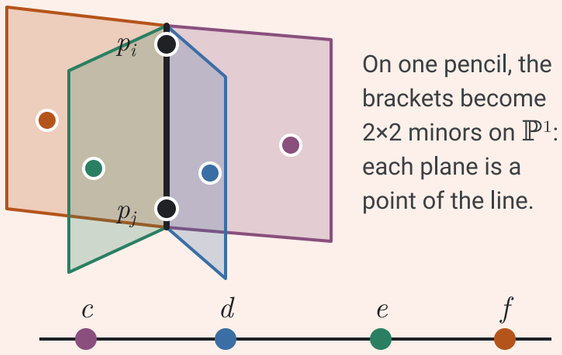
$$\mathrm{CR}_{ij}(c,d;e,f) = \frac{[ijce]\,[ijdf]}{[ijcf]\,[ijde]}$$

Unchanged by every gauge, sign included: $\det\ A$ and each λ_k cancel between the numerator and the denominator. For six points, three of them are the coordinates J:

$$J_1 = \mathrm{CR}_{12}(3,\,6;\,4,\,5)$$
$$J_2 = \mathrm{CR}_{34}(1,\,2;\,5,\,6)$$
$$J_3 = \mathrm{CR}_{56}(1,\,4;\,2,\,3)$$

Photo: four triangles share the edge $p_i p_j$ (white).

b Plücker relations



On one pencil, the brackets become 2×2 minors on $\mathbb{P}^1$: each plane is a point of the line.

$$GP_{ab} = [abcf]\,[abde]\,(1 - \lambda_1 - \lambda_2) \equiv 0$$

$$\lambda + (1 - \lambda) = 1$$

Each of the 15 relations is this identity on one pencil. So the residual ρ built from them is **identically zero on one reconstruction.**

Consistency must be judged *between* two reconstructions of the same points.

c Completeness

The $4n$ coordinates of n points split into the gauge orbit and what the pencils see.

$n = 12$: $48 = 27 + 21$

gauge $16 + n - 1$	$\mathbf{3n - 15}$

$$\operatorname{rank}\ J(\log|\mathrm{CR}_t|)_t = 3n - 15$$

Exact computation modulo $p = 2^{61} - 1$ for $n = 6,\ 7,\ 8,\ 12$ (ranks $3,\ 6,\ 9,\ 21$).

Near a generic configuration, equal pencil cross-ratios ⇔ the same points up to the gauge. The family detects every change that is not a change of frame.

Six points carry only three numbers: J_1, J_2, J_3.

d Test and conditioning

Consistency statistic of two measurements:

$$S = Q_{0.1}(\{D_t\}) = (r\ /\ \delta)\ Z + \mathrm{O}(r^2)$$

One threshold serves every noise scale. The low quantile follows the harmonic mean H of the per-pencil scales:

$$x_p\ /\ (p\ \sqrt{(\pi/2)}\ \ H) \to 1$$

Conditioning of the coordinates J:

$$\kappa_J = \max_k\ \|\nabla\ \log|J_k|\|$$

$$0.674\ \ \sigma\kappa_J \le \mathrm{med}(M) \le 1.383\ \ \sigma\kappa_J$$

$$\kappa_J \le 8\ /\ \ \delta_J \le 2\sqrt{(2L(S))}$$

First order ($\sigma\kappa_J \ll 1$). Tuple selection is maximum coverage: greedy reaches $\ge (1 - 1/e)\ \mathrm{OPT}$.

Status of each claim

Proved: gauge invariance, J as pencil cross-ratios, the Plücker identity, $\rho \equiv 0$, the scale law, the harmonic-mean limit, the κ_J bounds, the codimension $m - 3$ of the coplanar strata.

Computer-assisted: the rank $3n - 15$ (exact, modulo a prime). **Checked:** every claim has a test that can fail (verify_pencil.py, test_math_lock.py). No novelty is claimed.